

Assignment 4: Cluster Analysis of Pharmaceutical Companies

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Purpose

The purpose of this assignment is to use the k-Means clustering algorithm to explore and understand the structure of the pharmaceutical industry by analyzing financial data from 21 firms.

1. Data Loading and Preparation

First, we will load the necessary libraries and import the data from the `Pharmaceuticals.csv` file.

```
# Load necessary Libraries
library(tidyverse)    # For data manipulation and visualization
library(cluster)      # For clustering algorithms
library(factoextra)   # For visualizing clustering results

# Load the dataset
pharma_data <- read.csv("Pharmaceuticals.csv")
```

a) Cluster Analysis and Justification of Choices

In this step, we will conduct the cluster analysis using only the numerical variables (1 to 9) and justify the choices made.

Data Pre-processing: Variable Selection and Scaling

As per the assignment directions, we will use the first nine numerical financial variables: `Market_Cap`, `Beta`, `PE_Ratio`, `ROE`, `ROA`, `Asset_Turnover`, `Leverage`, `Rev_Growth`, and `Net_Profit_Margin`.

Justification for Scaling: The k-Means algorithm is based on Euclidean distance, which is sensitive to the scale of variables. For instance, `Market_Cap` (in billions) has a much larger scale than `Beta`. If we do not scale the data, variables with larger scales will disproportionately influence the clustering result. Therefore, we will **standardize** the data (to have a mean of 0 and a standard deviation of 1) to ensure all variables contribute equally to the analysis.

```
# Select only the numerical variables for clustering (columns 3 to 11)
pharma_numeric <- pharma_data[, 3:11]

# Set company names as row names for easier identification
rownames(pharma_numeric) <- pharma_data$Name

# Scale (standardize) the data
pharma_scaled <- scale(pharma_numeric)

# View the first few rows of the scaled data
head(pharma_scaled)
```

```
##           Market_Cap      Beta  PE_Ratio      ROE      ROA
## Abbott Laboratories  0.1840960 -0.80125356 -0.04671323  0.04009035  0.2416121
## Allergan, Inc.      -0.8544181 -0.45070513  3.49706911 -0.85483986 -0.9422871
## Amersham plc        -0.8762600 -0.25595600 -0.29195768 -0.72225761 -0.5100700
## AstraZeneca PLC     0.1702742 -0.02225704 -0.24290879  0.10638147  0.9181259
## Aventis             -0.1790256 -0.80125356 -0.32874435 -0.26484883 -0.5664461
## Bayer AG            -0.6953818  2.27578267  0.14948233 -1.45146000 -1.7127612
##           Asset_Turnover  Leverage Rev_Growth Net_Profit_Margin
## Abbott Laboratories    0.0000000 -0.2120979 -0.5277675      0.06168225
## Allergan, Inc.         0.9225312  0.0182843 -0.3811391     -1.55366706
## Amersham plc           0.9225312 -0.4040831 -0.5721181     -0.68503583
## AstraZeneca PLC        0.9225312 -0.7496565  0.1474473      0.35122600
## Aventis                -0.4612656 -0.3144900  1.2163867     -0.42597037
## Bayer AG               -0.4612656 -0.7496565 -1.4971443     -1.99560225
```

Determining the Optimal Number of Clusters (k)

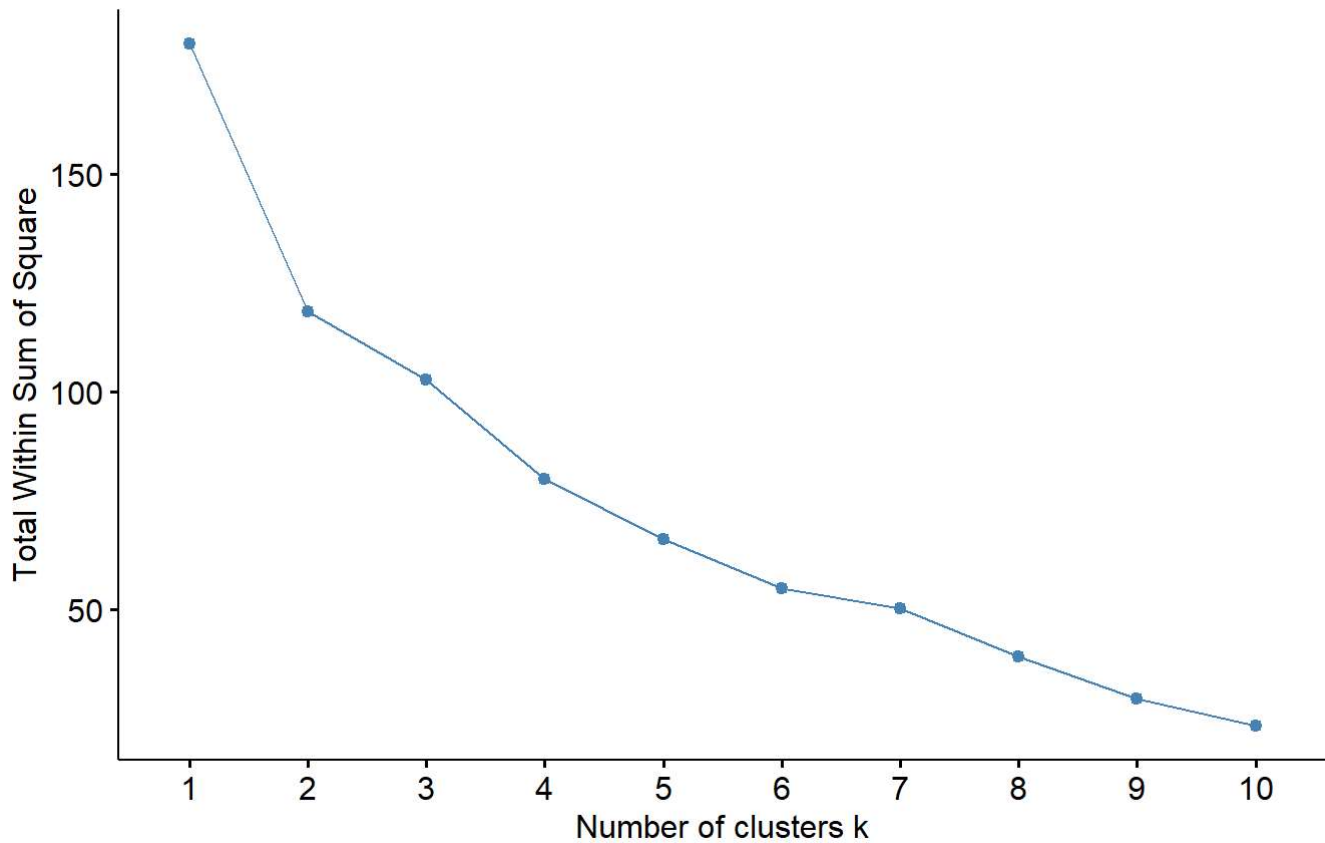
To choose an appropriate number of clusters, k , we will use two data-driven methods: the **Elbow Method** and the **Average Silhouette Method**.

Justification for Choice of k : These methods provide a systematic way to determine the optimal k by evaluating the quality of the clustering for different values of k . This avoids arbitrary selection and leads to more meaningful clusters.

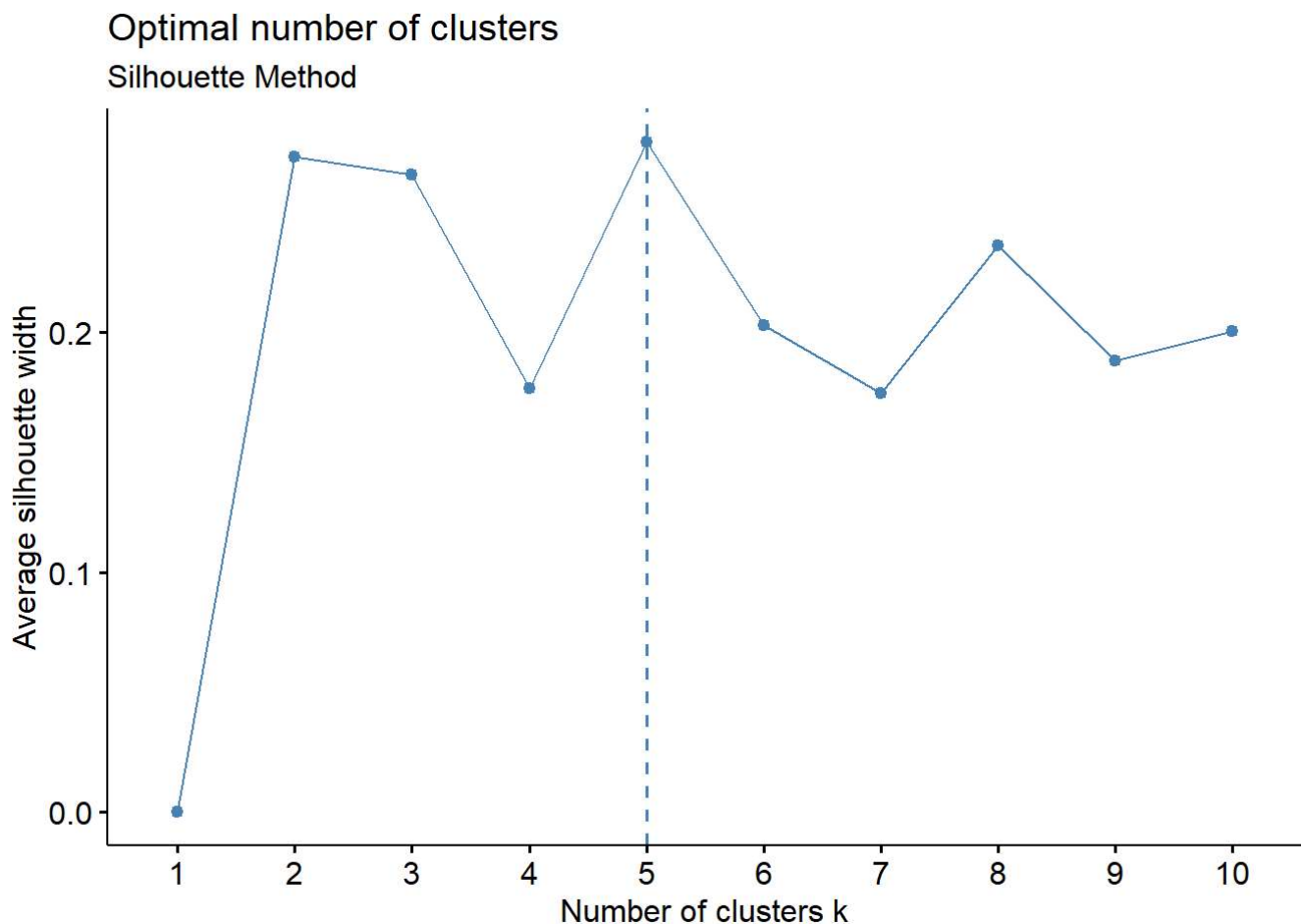
```
# Determine optimal k using the Elbow Method
fviz_nbclust(pharma_scaled, kmeans, method = "wss") +
  labs(subtitle = "Elbow Method")
```

Optimal number of clusters

Elbow Method



```
# Determine optimal k using the Silhouette Method  
fviz_nbclust(pharma_scaled, kmeans, method = "silhouette") +  
  labs(subtitle = "Silhouette Method")
```



Observation: The **Elbow Method** shows a distinct “elbow” at **k=3**, after which the decrease in the total within-cluster sum of squares (WSS) becomes less pronounced.

The **Silhouette Method** indicates **k=5** yields the highest average silhouette width, **but k=3** also shows a strong silhouette value and is supported by the Elbow Method and suitability for a small dataset.

Performing k-Means Clustering

Now, we will run the k-Means algorithm with $k=3$. We use `set.seed` for reproducibility and `nstart = 25` to ensure the algorithm finds a stable and optimal solution.

```
# Set a seed for reproducibility
set.seed(123)

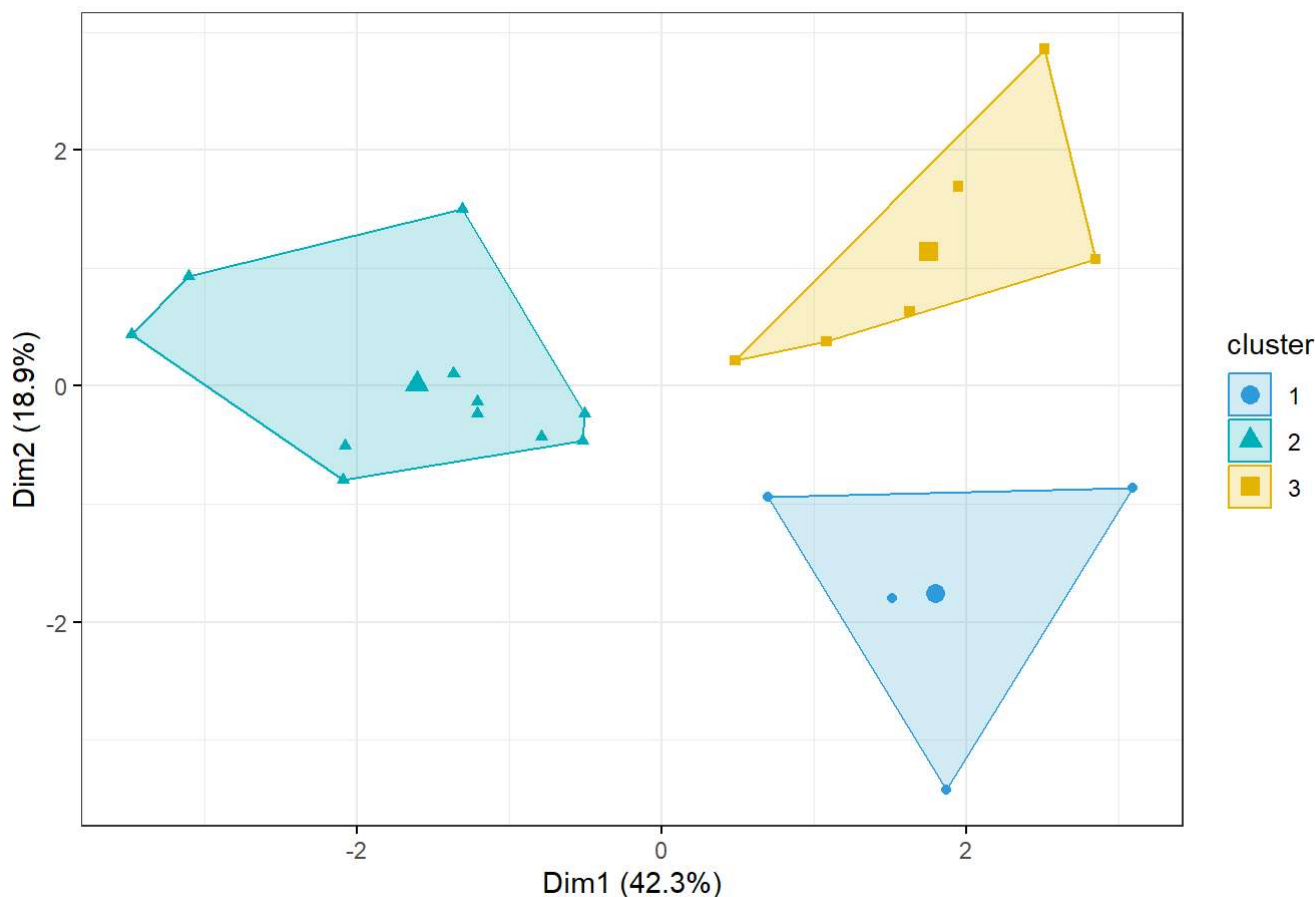
# Perform k-Means clustering with k=3
final_clusters <- kmeans(pharma_scaled, centers = 3, nstart = 25)

# Print the clustering results
print(final_clusters)
```

```
## K-means clustering with 3 clusters of sizes 4, 11, 6
##
## Cluster means:
##   Market_Cap      Beta    PE_Ratio      ROE      ROA Asset_Turnover
## 1 -0.6125361  0.2698666  1.3143935 -0.9609057 -1.0174553    0.2306328
## 2  0.6733825 -0.3586419 -0.2763512  0.6565978  0.8344159    0.4612656
## 3 -0.8261772  0.4775991 -0.3696184 -0.5631589 -0.8514589   -0.9994088
##   Leverage Rev_Growth Net_Profit_Margin
## 1 -0.3592866 -0.5757385    -1.3784169
## 2 -0.3331068 -0.2902163     0.6823310
## 3  0.8502201  0.9158889    -0.3319956
##
## Clustering vector:
##           Abbott Laboratories           Allergan, Inc.
##                   2                   1
##           Amersham plc           AstraZeneca PLC
##                   1                   2
##           Aventis           Bayer AG
##                   3                   1
## Bristol-Myers Squibb Company           Chattem, Inc
##                   2                   3
##           Elan Corporation, plc           Eli Lilly and Company
##                   3                   2
##           GlaxoSmithKline plc           IVAX Corporation
##                   2                   3
##           Johnson & Johnson Medicis Pharmaceutical Corporation
##                   2                   3
##           Merck & Co., Inc.           Novartis AG
##                   2                   2
##           Pfizer Inc           Pharmacia Corporation
##                   2                   1
##           Schering-Plough Corporation           Watson Pharmaceuticals, Inc.
##                   2                   3
##           Wyeth
##                   2
##
## Within cluster sum of squares by cluster:
## [1] 20.54199 43.30886 32.14336
## (between_SS / total_SS = 46.7 %)
##
## Available components:
##
## [1] "cluster"      "centers"      "totss"        "withinss"     "tot.withinss"
## [6] "betweenss"    "size"         "iter"         "ifault"
```

```
# Visualize the clusters
fviz_cluster(final_clusters, data = pharma_scaled,
              palette = c("#2E9FDF", "#00AFBB", "#E7B800"),
              geom = "point",
              ellipse.type = "convex",
              ggtheme = theme_bw()
)
```

Cluster plot



b) Interpretation of the Clusters

To interpret the clusters, we will analyze the mean values of the original (unscaled) numerical variables for each cluster. This allows us to understand the real-world characteristics of each group.

```
# Add the cluster assignment to the original numeric data
pharma_numeric_clustered <- pharma_numeric %>%
  mutate(Cluster = final_clusters$cluster)

# Calculate the mean of each variable for each cluster
cluster_summary <- pharma_numeric_clustered %>%
  group_by(Cluster) %>%
  summarise_all(mean)

# Print the summary table
print.data.frame(cluster_summary)
```

##	Cluster	Market_Cap	Beta	PE_Ratio	ROE	ROA	Asset_Turnover	Leverage
## 1	1	21.75500	0.5950000	46.90000	11.3	5.100000	0.7500000	0.3050000
## 2	2	97.11364	0.4336364	20.95455	35.7	14.954545	0.8000000	0.3254545
## 3	3	9.23500	0.6483333	19.43333	17.3	5.983333	0.4833333	1.2500000

##	Rev_Growth	Net_Profit_Margin
## 1	7.01000	6.65000
## 2	10.16455	20.17273
## 3	23.49000	13.51667

Cluster Interpretations (Corrected):

Based on the summary table above, we can interpret the clusters as follows:

- **Cluster 1: Stable, Highly-Valued Competitors.** These firms exhibit moderate `Market_Cap` (\$21.8B) and `ROE` (11.3). Their most defining feature is the **highest average PE_Ratio (46.9)**, indicating strong investor confidence in their future, despite having the lowest revenue growth and net profit margin in this dataset.
- **Cluster 2: Large, Profitable Blue-Chip Firms.** This group represents the industry giants, with the **highest average Market_Cap (\$97.1B)**, the **highest ROE (35.7)**, and the **highest Net_Profit_Margin (20.2%)**. Their financial performance is exceptionally strong and stable.
- **Cluster 3: High-Growth, High-Risk Emerging Firms.** This cluster contains the smallest firms, with the **lowest average Market_Cap (\$9.2B)**. However, they compensate with the **highest Revenue Growth (23.5%)**. Their high-risk nature is evident from the **highest average Leverage (1.25)**.

c) Pattern with Respect to Non-clustering Variables

Now, we examine if there is a pattern in the clusters with respect to the categorical variables (10-12) that were not used in forming the clusters.

```
# Add cluster assignments to the categorical variables
pharma_full_clustered <- pharma_data %>%
  select(Name, Median_Recommendation, Location, Exchange) %>%
  mutate(Cluster = final_clusters$cluster)
```

```
# Pattern by Median Recommendation
cat("Cluster distribution by Median Recommendation:\n")
```

```
## Cluster distribution by Median Recommendation:
```

```
table(pharma_full_clustered$Cluster, pharma_full_clustered$Median_Recommendation)
```

```
##
##      Hold Moderate Buy Moderate Sell Strong Buy
##    1      2          1          0          1
##    2      6          3          2          0
##    3      1          3          2          0
```

```
# Pattern by Location
cat("\nCluster distribution by Location:\n")
```

```
##
## Cluster distribution by Location:
```

```
table(pharma_full_clustered$Cluster, pharma_full_clustered$Location)
```

```
##
##      CANADA FRANCE GERMANY IRELAND SWITZERLAND UK US
##    1      1      0      1      0          0 1 1
##    2      0      0      0      0          1 2 8
##    3      0      1      0      1          0 0 4
```

```
# Pattern by Stock Exchange
cat("\nCluster distribution by Stock Exchange:\n")
```

```
##
## Cluster distribution by Stock Exchange:
```

```
table(pharma_full_clustered$Cluster, pharma_full_clustered$Exchange)
```

```
##
##      AMEX  NASDAQ  NYSE
##  1      0       0    4
##  2      0       0   11
##  3      1       1    4
```

Observations:

- **Median Recommendation:** Cluster 2 (Large Blue-Chips) firms are mostly rated as `Hold` or `Moderate Buy`. The other two clusters have more mixed recommendations.
- **Location:** Cluster 2 is dominated by firms headquartered in the **US** and **UK**. Most firms in Cluster 3 (High-Growth) are based in the **US**. Cluster 1 (Stable Competitors) is more geographically diverse.
- **Exchange:** Almost all firms in Cluster 1 and 2 are listed on the **NYSE**. The firms in Cluster 3 are listed on various exchanges, which is consistent with smaller, emerging companies.

d) Provide an Appropriate Name for Each Cluster

Based on the combined financial and categorical analysis, we can provide the following descriptive names for each cluster:

- **Cluster 1: Stable International Competitors**
 - **Reason:** This group consists of well-established, geographically diverse firms with stable performance. Their high PE ratios indicate strong investor confidence in their long-term value, even with lower current profitability.
- **Cluster 2: Global Pharma Giants**
 - **Reason:** This group includes companies with massive market caps, superior profitability, and a dominant presence in major markets (US/UK) and on the premier stock exchange (NYSE).
- **Cluster 3: Dynamic Growth Seekers**
 - **Reason:** These are smaller, riskier firms characterized by their aggressive revenue growth. They are dynamic and likely focused on innovation or niche markets to challenge the established giants.